

Fetch CAS summary

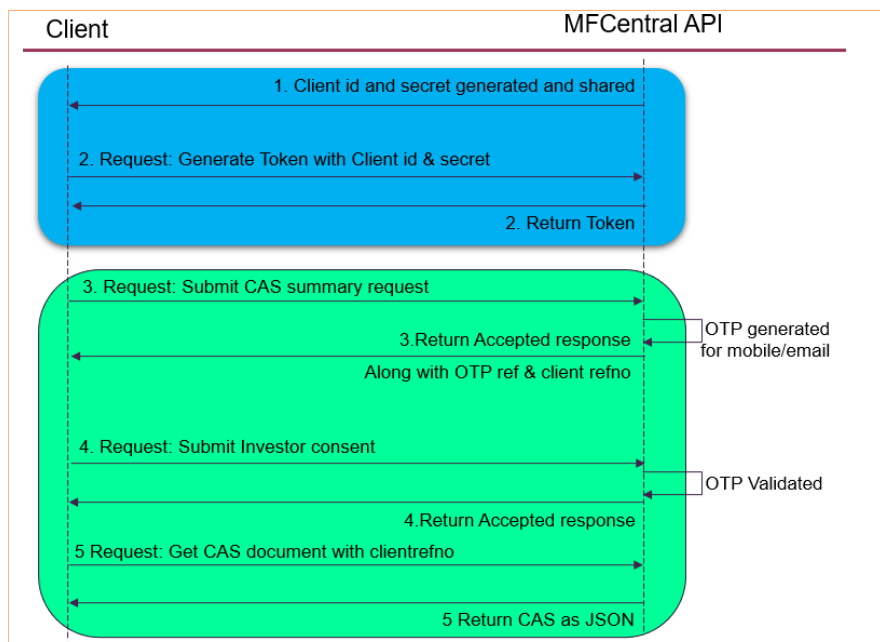
Description: This initial section is a pre-requisite for API integration to MFC.

1. What is MFcentral?

MFcentral is application that allows users to seamlessly navigate, search, analyze, and invest in MF tradingplatform for all funds serviced by RTA's. It lets you execute orders in real time, manage user portfoliosand place any Non-Financial and Financial Transactions request across all Mutual Funds.

2. Getting Started

This application provides a single unified portal for the user to access information regarding investments made across mutual funds. This Walkthrough goes over the basics of the authentication process as it applies to the MFCentral API using Json Web Tokens (JWT).



Above illustration explained w.r.t GetCASDocument API, the invoking process remains the same for other APIs too.

3. Onboarding & API Invoking Process

To access MFcentral APIs, you will need the client Id and Client secret to generate the token.

You may contact an MFcentral representative for generating Client Id and secret.

Once you have the Id and secret, connecting to our API is easy. You can proceed with generating the token, which should be passed in the header. Please refer to section 3.1

The following steps walk you through the basics to get started using the API.

3.1 Get Client ID and Client Secret.

You can get the client ID and secret, through an Offline process.

If you think the key has been compromised, contact immediately to generate a new Client Secret

Below details will be shared by MFCentral during on-boarding

```
String clientId = "";
String clientSecret = "";
String userName = "";
String password = "";
String encryptionDecryptionKey = "";
String privateKey = "";
String publicKey = "";
```

3.2 Generate Token

POST <baseUrl>/oauth2/token. This endpoint is accessible only over HTTPS. Plain HTTP connections are refused.

To generate a token that will be used for authorizing all subsequent APIs pass the Client ID, Client Secret and in the Authorization header.

Token generation involves passing Headers and Body together to the URL. Refer to the below Postman screenshot on how to generate token.

Every token object will have an expiry time. Ensure that you are creating a new token object if the existing token object is expired.(token expires in 30 days)

POST

https://uat.services.mfcentral.com/oauth/token

Params

Authorization

Headers (2)

Body

Pre-request Script

Tests

Key	Value	Description
☒ Content-Type	application/x-www-form-urlencoded	
☒ Authorization	Basic Z29hbHRlbGxlcjpwQjNjOTIEyTfc5QjZIR...	
New key	Value	Description

POST

https://uat.services.mfcentral.com/oauth/token

Params

form-data

☒ x-www-form-urlencoded

raw

binary

Key	Value	Description
☒ username	BCORK5NR39BT5VJ5D57H	
☒ password	5XGW4CVRAQ95KA3STOZ1	
☒ grant_type	password	
New key	Value	Description

Body

Cookies

Headers (11)

Test Results

Pretty

Raw

Preview

JSON

```

1 {
2   "access_token": "7ff2489a-e7b9-414a-b270-fe229043a9db",
3   "token_type": "bearer",
4   "refresh_token": "00a1fdae-685c-4bc6-8b9c-75b7055884ba",
5   "expires_in": 2591999,
6   "scope": "read write trust"

```

Sample:

Endpoint: <https://uatservices.mfcentral.com/oauth/token>

Headers:

Content-Type: application/x-www-form-urlencoded Authorization:
Basic dGVzdC1jbGllbnQxOnNzcGwkMTIzNA==

Body:

username: mfc-client1
password: Cams\$1234
grant_type: password

Response:

```
{
  "access_token": "dd33510a-f161-4818-8d35-65a64f617516", "token_type":
  "bearer",
  "refresh_token": "a0f749fa-2f97-44fd-be93-a5da24a22ca0", "expires_in":
  2591999,
  "scope": "read write trust"
}
```

For JAVA users - // Request 1 - Token Generation

```
String tokenGenerationUrl = "https://uatservices.mfcentral.com/oauth/token";
String token = getToken(tokenGenerationUrl, clientId, clientSecret,
userName,password);
```

3.3 Submitting Request to MFC

The token is sent in the header (Authorization Tag) along with the keyword “Bearer”. This token is authenticated and on success the validation of the signature process begins. If authentication fails, the response is returned, and the process is terminated.

Sample: URL: <https://uat.services.mfcentral.com/api/client/V1/<endpoint>>

3.3.1) Request Before encryption:

```
{
  "clientRefNo": "testing12345",
  "reqId": "083635320",
  "email": "",
  "mobile": "+918015933245",
  "pan": "EEAPK9544D",
  "pekrrn": "",
  "
}
```

Note: *clientRefNo* should be unique across different unique requests from client for tracking purpose

3.3.2) Encrypt the above request:

The above request should be encrypted using SHA 256 hash logic

To encrypt/decrypt data, use the below details.

During Encryption: Final value should be base 64 encoded value from the encrypted result.

Take SHA-256 hash of the shared key and use the first 32 byte of the hash as your secret key for both encryption and decryption.

Algorithm	:	AES Symmetric 256-bit
Cipher Mode	:	CBC
Padding	:	PKCS5Padding
IV text	:	<Constant> - will be shared via email separately

Reference

Use `CryptUtils.encryptDecrypt()` for decryption

Sample Usage for Encryption :`CryptUtils.encryptDecrypt("ENCRYPT", payload, encryptionKey);`

3.3.3) Generate Signature for the above encrypted request:

Generate a signature which should be passed along with the request. The signed value is a JWS (JSON WEB SIGNATURE) based value. Use the below detail to generate a JWS.

Algorithm: RS256 (RSA using SHA-256 hash algorithm)

Serialization: Compact Serialization, detached payload

Header (base64url-encode payload): false

Reference :

Use generateSignature function

Sample Usage: generateSignature(encryptedText, privateKey);

```
public static String generateSignature(String payload, String privateKey) {
    JSONObject output = new JSONObject();

    try {
        RsaJsonWebKey rsaJsonWebKey = getPrivateKey(privateKey);
        JsonWebSignature jws = new JsonWebSignature();
        jws.setAlgorithmHeaderValue(AlgorithmIdentifiers.RSA_USING_SHA256);
        jws.setKeyIdHeaderValue(rsaJsonWebKey.getKeyId());
        jws.getHeaders().setObjectHeaderValue(HeaderParameterNames.BASE64URL_ENCODE_PAYLOAD, false);
        jws.setPayload(payload);
        jws.setKey(rsaJsonWebKey.getPrivateKey());

        String digitalSignature = jws.getDetachedContentCompactSerialization();

        output = new JSONObject();
        output.put("request", payload);
        output.put("signature", digitalSignature);

        return output.toString();
    } catch (Exception ex) {
        ex.printStackTrace();
    }

    return null;
}
```

```

public static RsaJsonWebKey getPrivateKey(String privateKey) throws Exception {
    JsonWebKeySet jsonWebKeySet = new JsonWebKeySet(privateKey);
    RsaJsonWebKey rsaJsonWebKey = (RsaJsonWebKey) jsonWebKeySet.getJsonWebKeys().get(0);
    return rsaJsonWebKey;
}

```

sample process – for JAVA based Users:

```

// Inputs to the request (Switch API structure)

String pan = "ACXXXX91XX";
String pekrn = "";
String mobile = "+91XXXXX70XX9";
String email = "";
String clientReferenceNumber = "testing12345"; //dynamic value generated by
client

// Request 2 - submit pan/pekrn & mobile/email to MFCentral for Financial
Transactions.

UAT URL : https://uat-services.mfcentral.com/api/client/V1/redeem

String request = Switch structure as given above

String redeemRequestUrl = "https://uat-services.mfcentral.com/api/client/V1/redeem";
JSONObject response = submitRequestToMFCentral(
    RequestUrl,
    token,
    clientId,
    encryptionDecryptionKey,
    privateKey, publicKey, request);

```

Response From MFC

```

{
    "reqId": 56475, //Generated by MFC
    "otpRef": "9b69c14e-76f5-4ccb-9cd1-22e20793c4e9", //Generated by MFC
    "userSubjectReference": "",
    "clientRefNo": "testing12345"
}

```

3.3.4) Submitting the request to MFCentral

Once you have generated the signature, submit your request in the below request format.

```
{ "signature": "<signature generated in step 3.3.3>",  
  "request": "<encrypted request generated in step 3.3.2>" }
```

Headers:

Content-Type: application/json

ClientId: test-client1

Authorization: Bearer dd33510a-f161-4818-8d35-65a64f617516

Method: POST

Reference

```
HttpHeaders headers = new HttpHeaders();  
headers.add("Content-Type", "application/json");  
headers.add("ClientId", clientId);  
headers.add("Authorization", "Bearer " + token);  
  
RestTemplate restTemplate = new RestTemplate();  
HttpEntity<String> request = new HttpEntity<String>(mfcRequest, headers);  
  
JSONObject responseJson = null;  
try {  
    ResponseEntity<String> response = restTemplate.postForEntity( url, request, String.class );  
  
    String responseBody = response.getBody();  
    responseJson = new JSONObject(responseBody);  
} catch (HttpStatusCodeException e) {  
  
    String responseBody = e.getResponseBodyAsString();  
    responseJson = new JSONObject(responseBody);  
}
```

Sample Response: { "signature": "", "response": "" }

3.3.5) Verifying the signature

MFCentral returns response payload along with the digital signature. Verify the signature before processing the response.

Reference:

Use verifySignature

function Sample Usage:

```
verifyDigitalSignature(response, signature, publicKey);
```

```
public static boolean verifyDigitalSignature(String payload, String digitalSignature, String publicKey) {  
    try {  
        RsaJsonWebKey rsaJsonWebKey = getPublicKey(publicKey);  
  
        JsonWebSignature verifierJws = new JsonWebSignature();  
        verifierJws.setAlgorithmConstraints(new AlgorithmConstraints(  
            AlgorithmConstraints.ConstraintType.WHITELIST,  
            AlgorithmIdentifiers.RSA_USING_SHA256));  
        verifierJws.setKey(rsaJsonWebKey.getPublicKey());  
        verifierJws.setCompactSerialization(digitalSignature);  
        verifierJws.setPayload(payload);  
        return verifierJws.verifySignature();  
    } catch (Exception ex) {  
        ex.printStackTrace();  
    }  
    return false;  
}
```

3.36) Decrypt the response

Before decryption, decode the base 64 value first.

Algorithm	:	AES Symmetric 256-bit
Cipher Mode	:	CBC
Padding	:	PKCS5Padding
IV text	:	<constant>

Reference

Use `CryptUtils.encryptDecrypt()` for decryption

Sample Usage:

```
CryptUtils.encryptDecrypt("DECRYPT", response, encryptionKey);
```

3.4 Submitting Investor consent to MFCentral (Validating OTP):

Upon successful authentication, you shall request for user OTP validation.

Once the User enters the OTP, MFcentral will validate and if validation is successful, shall allow to proceed with the next step, else return error as OTP validation failed.

For every request OTP validation shall be a mandatory process.

Below is the process for Investor consent

URL: <https://uat.services.mfcentral.com/api/client/V1/investorconsent>

Headers:

Content-Type: application/json

ClientId: test-client1

Authorization: Bearer dd33510a-f161-4818-8d35-65a64f617516

3.4.1) Request Before encryption:

```
{
    "reqId": 28364,
    "otpRef": "83057769-d7dd-4636-83e7-a5a898066f73",
    "userSubjectReference": "",
    "clientRefNo": "client1-123456",
    "enteredOtp": "973947"
}
```

reqId – generated from response – step 3.3.6

otpRef – generated from response – step 3.3.6

perform steps 3.3.2 to 3.3.6 for the payload to be sent and to get the response

```
// Request 3 - Submit investor Consent to MFCentral  
  
String otpValidationUrl =  
    "https://uat-services.mfcentral.com/api/client/V1/investorconsent";  
JSONObject otpRequest = new  
    JSONObject("{\"reqId\":\"55718\",\"otpRef\":\"35979fc2-e2c8-4faa-b6dc-ab6505a4612e\", \"userSubjectReference\":\"\", \"clientRefNo\":\"testing12345\"}");  
otpRequest.put("enteredOtp", "574224"); // otp entered by user
```

```
submitRequestToMFCentral(  
    otpValidationUrl, token, clientId, encryptionDecryptionKey,  
    private key, public key, otpRequest.ToString())
```

Response:

Response from MFCentral: Accepted status 202 <<empty>>

3.5 Submit request to get the status of the transaction:

Upon successful OTP validation, you can invoke the MFcentral APIs. The API response structure are provided in section 4

Below sample is provided for redemption, likewise you may use the respective service endpoints.

Endpoint: <https://sit-services.mfcentral.com/api/client/V1/getFtTransactionStatus>

Headers:

Content-Type: application/json

ClientId: test-client1

Authorization: Bearer dd33510a-f161-4818-8d35-65a64f617516

3.5.1) Request Before encryption:

Use the reqId and clientRefNo sent in the previous request 3.4.1

```
{"reqId":28364,"clientRefNo":"testing-12345"}
```

perform steps 3.3.2 to 3.3.6 for the payload to be sent and get the response

```
// Request 4 - Get Transaction Status  
  
String transactionstatus =  
    "https://sit-services.mfcentral.com/api/client/V1/getFtTransactionStatus";
```

Note: As this is asynchronous call, you may invoke the API to get the actual status of the Transaction.

There is no limit/restriction for the no of hits for Transaction status API

API Flow:

1. Encrypt JSON Request(<https://uat.services.mfcentral.com/api/test/encrypt>)
2. Generate Signature for calling CAS API(<https://uat.services.mfcentral.com/api/test/generateSignature>)
3. Invoke the CAS Summary API: <https://uat.services.mfcentral.com/api/client/V1/submitcassummaryrequest>
4. Decrypt the Response Received : (<https://uat.services.mfcentral.com/api/test/decrypt>)
5. Encrypt the request for Investor Consent : <https://uat.services.mfcentral.com/api/test/encrypt>
6. Generate Signature(<https://uat.services.mfcentral.com/api/test/generateSignature>)
7. Invoke Investor Consent API(<https://uat.services.mfcentral.com/api/client/V1/investorconsent>)
8. Encrypt the request for CAS Document API : <https://uat.services.mfcentral.com/api/test/encrypt>
9. Generate Signature(<https://uat.services.mfcentral.com/api/test/generateSignature>)
10. Invoke CAS Document API (<https://uat.services.mfcentral.com/api/client/V1/getcasdocument>)
11. Decrypt to get CAS : (<https://uat.services.mfcentral.com/api/test/decrypt>)

4. CAS

Description: A common account statement is a consolidated statement where you can view all your mutual fund investments and their statuses, irrespective of the fund, RTA or depository where your investments are held

This post request is designed to receive switch transaction requests and process them in accordance with RTA rules and validations.

Request endpoints:

CAS request summary api : <https://uat.services.mfcentral.com/api/client/V1/submitcassummaryrequest>

CAS document api: <https://uat.services.mfcentral.com/api/client/V1/getcasdocument>

Header: Instead of individual key-value pairs, we're providing bulk edit content for direct copying and pasting during Postman testing. Please follow the steps below:

1. Navigate to the header section.
2. Select the "Bulk Edit" option.
3. Paste the provided content into the designated area.

Headers  Hide auto-generated headers

```
Content-Type:application/json
Authorization:Bearer {{token}}
origin:https://uatapp.mfcentral.com
ClientId:ranjani
```

```
Content-Type:application/json
Authorization:Bearer {{token}}
origin:https://uatapp.mfcentral.com
ClientId:ranjani
```

Parameters

Parameter	Data type(Length)	Mandatory	Description	Validation
clientRefNo	String(30 characters)	REQUIRED	A client reference number serves as a unique key for each API call, facilitating the updating of the internal database for logging requests	Ensure that the client reference number is a unique alphanumeric or numeric value within 30 characters
reqId	String(50)	OPTIONAL	The request id is a series of characters or numbers used to identify each request received for processing. Pass this value as empty or NULL	
email	String(500)	CONDITIONAL REQUIRED	Email address of the investor registered in the PAN/folio number. Either email or mobile number is mandatory, Do not pass both values	A non-empty string containing "@" symbol and should have atleast one '.'
mobile	String(30)	CONDITIONAL REQUIRED	Mobile number of the investor registered in the PAN/folio number. Either email or mobile number is mandatory, Do not pass both values	Should be in the format "+91" followed by 10 digits or simply 10 digits
pan	String(45)	CONDITIONAL REQUIRED	PAN (Permanent Account Number) of the investor. Either PAN or PEKRN is mandatory. Do not pass both values	Should follow the format of 5 alpha characters, followed by 4 numeric characters, and ending with 1 alpha character.
pekrn	String(15)	CONDITIONAL REQUIRED	PEKRN is a unique identification number provided to individuals who are exempt from providing a Permanent Account Number (PAN) during the Know Your Customer (KYC) process for mutual fund investments. Either PAN or PEKRN is mandatory. Do not pass both values	

Request

```
{
  "clientRefNo": "2492dcc241753123",
  "reqId": "1101",
  "pan": "ALNPR6026P",
  "mobile": "9884174067",
  "email": "",
  "pekrrn": ""
}
```

RESPONSE:

Invokes the investor consent api by triggering an OTP

Response from MFCentral: Accepted status 200<<empty>>(if otp matches with the otp ref)

CAS Document api:

Parameters

Parameter	Data type(Length)	Mandatory	Description	Validation
clientRefNo	String(30 characters)	REQUIRED	A client reference number serves as a unique key for each API call, facilitating the updating of the internal database for logging requests	Ensure that the client reference number is a unique alphanumeric or numeric value within 30 characters
reqId	String(50)	REQUIRED	The request id is a series of characters or numbers used to identify each request received for processing. Pass this value as empty or NULL	

Request

```
{
  "reqId": 2563565,
  "clientRefNo": "2492dcc241753123"
}
```

Response

```
{
  "reqId": "2563566",
  "pan": "ALNPR6026P",
  "pekrrn": "",
  "mobile": "9884174067",
  "email": "",
  "data": [
    {
```

```
"summary": [
  {
    "amc": "H",
    "amcName": "MFHMF Test Fund",
    "isDemat": "N",
    "currentMktValue": "14088984.96",
    "costValue": "11550590.64",
    "gainLoss": "2538394.32",
    "gainLossPercentage": "21.98"
  }
],
"schemes": [
  {
    "amc": "H",
    "amcName": "MFHMF Test Fund",
    "folio": "10006292",
    "investorName": "GRAZSUTPSGEC",
    "age": 68,
    "mobile": "9884174067",
    "email": "ranjani.sr@camsonline.com",
    "taxStatus": "01",
    "modeOfHolding": "Single",
    "transactionSource": "Channel",
    "schemeCode": "BFG",
    "schemeName": "HDFC Balanced Fund - Regular Plan - Growth ",
    "idcwChangeAllowed": "False",
    "schemeOption": "Growth",
    "assetType": "EQUITY",
    "schemeType": "Balanced",
    "nav": "147.069",
    "navDate": "01-Jun-2018",
    "closingBalance": "0.00",
    "availableUnits": "0.000",
    "availableAmount": "0.00",
    "currentMktValue": "0.00",
    "costValue": "0.00",
    "gainLoss": "0.00",
    "gainLossPercentage": "0.00",
    "isDemat": "N",
    "lienUnitsFlag": "N",
    "decimalUnits": 3,
    "decimalAmount": 2,
    "decimalNav": 3,
    "brokerCode": "ARN-111569",
    "brokerName": "ANAND RATHI WEALTH LIMITED",
    "isin": "INF179K01392",
    "purAllow": "N",
    "redAllow": "N",
    "swtAllow": "N",
    "sipAllow": "N",
    "stpAllow": "N",
    "swpAllow": "N",
    "planMode": "R",
    "dpId": "",
    "mobileRelationship": "",
    "emailRelationship": "SE",
```

```

        "newFolio": "N",
        "nomineeStatus": "Y",
        "lienEligibleUnits": "0.000",
        "bank": {
            "accountNo": "00271930014183",
            "accountType": "SB",
            "name": "HDFC Bank Ltd",
            "branch": "Greater Kailash 2",
            "city": "New Delhi",
            "pincode": "",
            "micr": "",
            "ifsc": "HDFC0000027",
            "neftifsc": "HDFC0000027"
        },
        "rtaNm": "CAMS"
    },
    "portfolio": [
        {
            "currentMktValue": "22631954.56",
            "costValue": "11749460.00",
            "gainLoss": "10882494.56",
            "gainLossPercentage": "92.62",
            "isDemat": "N"
        },
        {
            "currentMktValue": "121884.12",
            "costValue": "70000.00",
            "gainLoss": "51884.12",
            "gainLossPercentage": "74.12",
            "isDemat": "Y"
        }
    ],
    "investorDetails": {
        "address": {
            "line1": "S/O GAJENDRA RAUT H.NO 3.-3-76-123/10/A ",
            "line2": "ENCLAVE JANPRIYA UTOPIA HYDERGUDA NEAR J",
            "line3": "TELANGANA",
            "state": "TELANGANA",
            "pincode": 500048,
            "country": "INDIA",
            "city": "ATTAPUR"
        },
        "email": "xxx@kfintech.com",
        "mobile": "9884174067",
        "investorName": "PRASANNA KUMAR RAUT"
    },
    "statementHoldingFilter": "NON-ZERO"
}

```

Parameter	Data type(Length)	Description
-----------	-------------------	-------------

reqId	String(50)	The request id is a series of characters or numbers used to identify each request received for processing. Pass this value as empty or NULL
email	String(500)	Email address of the investor registered in the PAN/folio number. Either email or mobile number is mandatory, Do not pass both values
mobile	String(30)	Mobile number of the investor registered in the PAN/folio number. Either email or mobile number is mandatory, Do not pass both values
pan	String(45)	PAN (Permanent Account Number) of the investor. Either PAN or PEKRN is mandatory. Do not pass both values
pekrn	String(15)	PEKRN is a unique identification number provided to individuals who are exempt from providing a Permanent Account Number (PAN) during the Know Your Customer (KYC) process for mutual fund investments. Either PAN or PEKRN is mandatory. Do not pass both values
amc	String(5)	AMC (Asset Management Company) code
amcName	String(100)	This provides the full name of the Asset Management Company
taxStatus	String	Tax status code for the investment (e.g., "01" might indicate resident individual).
modeOfHolding	String	How the investment is held (e.g., "Single" for sole ownership).
transactionSource	String	Channel through which the investment was made (e.g., "Channel" for bank or broker).
schemeCode	String	Unique code assigned to the specific mutual fund scheme.
schemeName	String	Full name of the mutual fund scheme you hold.
idcwChangeAllowed	String (Boolean)	Indicates if switching between direct & regular plans (IDCW) is allowed (e.g., "False" for not allowed).
schemeOption	String	Reinvestment option (e.g., "Growth" for dividends reinvested).
assetType	String	Primary underlying investment of the scheme (e.g., "EQUITY" for stocks).
schemeType	String	Investment style of the scheme (e.g., "Balanced" for a mix of equity and debt).
nav	Decimal	Net Asset Value per unit of the scheme on a specific date.
navDate	Date	Date for which the Net Asset Value (NAV) is provided.
closingBalance	Decimal	Number of units held in the scheme at the end of the period (e.g., "0.00" for zero units).
isDemat	String(1)	This indicates whether the holding is in dematerialized form or physical
currentMktValue	Integer	This represents the current market value of your holding in the mutual fund scheme. It is likely the closing price of the scheme on a specific date.
costValue	Integer	This indicates the total cost price at which you purchased all your units in this scheme.
gainLoss	String	This shows the difference between the current market value and the cost value. A positive value indicates unrealized capital gain, while negative indicates unrealized capital loss.
gainLossPercentage	String	This represents the gain or loss as a percentage of the cost value
isDemat	String (Boolean)	This signifies whether the holding is in dematerialized form (electronic or physical)
lienUnitsFlag	String (Boolean)	This could indicate if any units in this holding are under lien (restricted for selling or transacting) due to a loan or other reason
decimalUnits	Integer	This might allow holding units with decimals
decimalAmount	Integer	Similar to decimalUnits, this might allow investment amounts with decimals
decimalNav	Integer	This could indicate that the NAV is represented with decimals
brokerCode	String	This might be included if the report is generated by a broker

		platform and shows the broker code associated with the holding.
brokerName	String	Similar to brokerCode, this could provide the name of the broker if applicable.
purAllow	String (Boolean)	This parameter indicates whether you can make additional purchases of units in the specific mutual fund scheme
redAllow	String (Boolean)	This refers to whether you are allowed to redeem (sell back) your units in a specific mutual fund scheme
swtAllow	String (Boolean)	This refers to whether you are allowed to switch your investment units from one mutual fund scheme to another within the same AMC
sipAllow	String (Boolean)	This shows whether Systematic Investment Plans (SIP) are allowed for this scheme
stpAllow	String (Boolean)	This parameter indicates whether Systematic Transfer Plans (STPs) are allowed for the specific mutual fund scheme.
swpAllow	String (Boolean)	This parameter indicates whether Systematic Withdrawal Plans (SWPs) are allowed for the specific mutual fund scheme.
planModeclosi	String	This could potentially indicate the investment plan mode which is direct or regular
dpId	String	DPID (Depository Participant Identification) A DP is an intermediary registered with a central depository, such as the National Securities Depository Limited (NSDL) or the Central Depository Services Limited (CDSL), which facilitates the holding, transfer, and trading of securities in electronic form.
mobileRelationship	String	This field lets you define how the phone number you provide is associated with your MF investment.
emailRelationship	String	This field allows you to categorize the email address you provide
newFolio	String (Boolean)	This parameter might be used during account creation or a new folio request (separate holding within an AMC). Its presence suggests the potential for generating a new portfolio.
nomineeStatus	String	This could indicate whether a nominee (someone you appoint to inherit your investments) is registered for this portfolio
lienAvailableUnits	Integer	It's the number of units held by the lender to take and hold or sell the property of a debtor as security or payment for a debt.the number of units under lien is usually restricted for selling due to a loan or other reason
bank.accountNo	Integer	This could be your bank account number associated with your mutual fund investments
bank.accountType	String	Specifies the type of bank account linked to the SIP, such as savings account (SB), current account (CA), etc.
bank.name	String	Indicates the name of the bank where the investment linked bank account is held
bank.branch	String	branch of the bank where the investment linked bank account is held
bank.city	String	Denotes the city where the bank branch associated with the SIP account is located
bank.pincode	Integer	Represents the postal code or ZIP code of the bank branch's location.
bank.micr	String	This could be the Magnetic Ink Character Recognition (MICR) code of your bank branch, used for cheque processing
bank.ifsc	String	Indian Financial System Code(IFSC), A unique alphanumeric code that is used for online fund transfer transactions done via NEFT, RTGS and IMPS
bank.neftifsc	String	The NEFT IFSC (National Electronic Funds Transfer Indian Financial System Code) is a unique alphanumeric code required for initiating NEFT (National Electronic Funds Transfer) transactions in India. This code uniquely identifies a specific branch of a bank participating in the NEFT network, enabling

		seamless electronic fund transfers between accounts held in different branches or banks across the country.
rtaNam e	String	This could be the name of the Record and Transfer Agent (RTA) associated with the AMC that manages your mutual fund scheme(s)
costValue	Integer	This indicates the total cost price at which you purchased all your units in this scheme.
gainLoss	String	This shows the difference between the current market value and the cost value. A positive value indicates unrealized capital gain, while negative indicates unrealized capital loss.
gainLossPercent age	String	This represents the gain or loss as a percentage of the cost value
isDemat	String (Boolean)	This signifies whether the holding is in dematerialized form (electronic or physical)
lienUnitsFlag	String (Boolean)	This could indicate if any units in this holding are under lien (restricted for selling or transacting) due to a loan or other reason
decimalUnits	Integer	This might allow holding units with decimals
decimalAmount	Integer	Similar to decimalUnits, this might allow investment amounts with decimals
decimalNav	Integer	This could indicate that the NAV is represented with decimals
brokerCode	String	This might be included if the report is generated by a broker platform and shows the broker code associated with the holding.
brokerName	String	Similar to brokerCode, this could provide the name of the broker if applicable.
purAllow	String (Boolean)	This parameter indicates whether you can make additional purchases of units in the specific mutual fund scheme
redAllow	String (Boolean)	This refers to whether you are allowed to redeem (sell back) your units in a specific mutual fund scheme
swtAllow	String (Boolean)	This refers to whether you are allowed to switch your investment units from one mutual fund scheme to another within the same AMC
sipAllow	String (Boolean)	This shows whether Systematic Investment Plans (SIP) are allowed for this scheme
stpAllow	String (Boolean)	This parameter indicates whether Systematic Transfer Plans (STPs) are allowed for the specific mutual fund scheme.
swpAllow	String (Boolean)	This parameter indicates whether Systematic Withdrawal Plans (SWPs) are allowed for the specific mutual fund scheme.
planMode	String	This could potentially indicate the investment plan mode which is direct or regular
dpId	String	DPID (Depository Participant Identification) A DP is an intermediary registered with a central depository, such as the National Securities Depository Limited (NSDL) or the Central Depository Services Limited (CDSL), which facilitates the holding, transfer, and trading of securities in electronic form.
mobileRelations hip	String	This field lets you define how the phone number you provide is associated with your MF investment.
emailRelationsh ip	String	This field allows you to categorize the email address you provide
newFolio	String (Boolean)	This parameter might be used during account creation or a new folio request (separate holding within an AMC). Its presence suggests the potential for generating a new portfolio.
nomineeStatus	String	This could indicate whether a nominee (someone you appoint to inherit your investments) is registered for this portfolio
lienAvailableUn its	Integer	It's the number of units held by the lender to take and hold or sell the property of a debtor as security or payment for a debt.the number of units under lien is usually restricted for selling due to a loan or other reason
bank.accountNo	Integer	This could be your bank account number associated with your

		mutual fund investments
bank.accountType	String	Specifies the type of bank account linked to the SIP, such as savings account (SB), current account (CA), etc.
bank.name	String	Indicates the name of the bank where the investment linked bank account is held
bank.branch	String	branch of the bank where the investment linked bank account is held
bank.city	String	Denotes the city where the bank branch associated with the SIP account is located
bank.pincode	Integer	Represents the postal code or ZIP code of the bank branch's location.
bank.micr	String	This could be the Magnetic Ink Character Recognition (MICR) code of your bank branch, used for cheque processing
bank.ifsc	String	Indian Financial System Code(IFSC), A unique alphanumeric code that is used for online fund transfer transactions done via NEFT, RTGS and IMPS
bank.neftifsc	String	The NEFT IFSC (National Electronic Funds Transfer Indian Financial System Code) is a unique alphanumeric code required for initiating NEFT (National Electronic Funds Transfer) transactions in India. This code uniquely identifies a specific branch of a bank participating in the NEFT network, enabling seamless electronic fund transfers between accounts held in different branches or banks across the country.
rtaName	String	This could be the name of the Record and Transfer Agent (RTA) associated with the AMC that manages your mutual fund scheme(s)
line1	String	Denotes the address line1 of the investor, providing a detailed postal address
line2	String	Address line 2
line3	String	Address line 3
state	String	Represents the state of the investor's address
country	String	Represents the country of the investor's address
investorName	String(50)	This parameter represents the name of the investor. It identifies the individual or entity making the investment.

ERROR:

A failure response is preceded by the corresponding 40x or 50 x HTTP headers. The code key in the response contains the value error. The message key contains a textual description of the error.

Below are the possible error codes:

Status Code	Actual Status	Reason/Description
400	Valid Business error	In this case, refer error description of the error & take action, these are generally business validation errors. Basis the error received, you can ask the user to remediate certain fields & resubmit the transaction again
500	Internal server error or Connectivity error	In this case, refer error description of the error & take action, these are generally technical errors.
401	Unauthorized	Invalid token or client Id

422	Un-processable Entity (WebDAV)	Invalid Input or Request Format
404	Not Found	The requested resource could not be found on the server. Please verify the endpoint URL and try again.
202	Accepted	In this case, the request has been accepted for processing, but the processing has not been completed.
200	Successful	Request has been accepted and it is a successful request

AMC CODE MASTER:

S. No.	AMC CODE	AMC NAME	AMC SHORT NAME
1	IF	360 ONE Mutual Fund	360 ONE MF
2	B	Aditya Birla Sun Life Mutual Fund	BIRLA MF
3	G	Bandhan Mutual Fund	Bandhan MF
4	D	DSP Mutual Fund	DSP MF
5	FTI	Franklin Templeton Mutual Fund	Franklin Templeton MF
6	H	HDFC Mutual Fund	HDFC MF
7	O	HSBC Mutual Fund	HSBC MF
8	P	ICICI Prudential Mutual Fund	I-PRU MF
9	K	Kotak Mutual Fund	KOTAK MF
10	MM	Mahindra Manulife Mutual Fund	MAHINDRA MF
11	PLF	NAVI Mutual Fund	NAVI MF
12	PP	PPFAS Mutual Fund	PPFAS MF
13	L	SBI Mutual Fund	SBI MF
14	SH	Shriram Mutual Fund	SHIRAM MF
15	T	Tata Mutual Fund	TATA MF
16	UK	Union Mutual Fund	UNION MF
17	Y	WhiteOak Capital Mutual Fund	WhiteOak Capital MF
18	128	AXIS MUTUAL FUND	Axis MF
19	178	BARODA BNP PARIBAS MUTUAL FUND	Baroda MF
20	116	BOI AXA MUTUAL FUND	Boi MF
21	189	Bajaj Finserv Mutual Fund	Bajaj Finserv
22	101	CANARA ROBECO MUTUAL FUND	Canara MF
23	118	EDELWEISS MUTUAL FUND	Edelweiss MF
24	125	Groww Mutual Fund	Groww MF
25	120	INVESCO MUTUAL FUND	Invesco MF
26	152	ITI MUTUAL FUND	Iti MF
27	105	JM FINANCIAL MUTUAL FUND	Jm MF
28	102	LIC MUTUAL FUND	Lic MF
29	117	MIRAE ASSET MUTUAL FUND	Mirae MF
30	127	MOTILAL OSWAL MUTUAL FUND	Motilal MF

31	RMF	NIPPON INDIA MUTUAL FUND	Nippon MF
32	187	NJ MUTUAL FUND	Nj MF
33	129	PGIM INDIA MUTUAL FUND	Pgim MF
34	166	QUANT MUTUAL FUND	Quant MF
35	123	QUANTUM MUTUAL FUND	Quantum MF
36	188	SAMCO MUTUAL FUND	Samco MF
37	176	SUNDARAM MUTUAL FUND	Sundaram MF
38	104	TAURUS MUTUAL FUND	Taurus MF
39	185	TRUST MUTUAL FUND	Trust MF
40	108	UTI MUTUAL FUND	UTI MF
41	139	OLD BRIDGE MUTUAL FUND	Old Bridge MF
42	HLS	HELIOS MUTUAL FUND	Helios MF
43	Z	ZERODHA MUTUAL FUND	Zerodha MF

Other Api master

Code	Description
R	Redemption
S	Switch
NP	New Purchase (Lumpsum / SIP)
AP	Additional Purchase (Lumpsum / SIP)
ST	STP
SW	SWP
SS	Summary (SIP / STP/ SWP)
SPC	Pause and Cancel submit
SPCV	Pause and Cancel Validate